SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

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CSA37- SOFTWARE TESTING

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Requirement Analysis and Negative Test Scenarios and Test Cases for a Railway Reservation System

Aim

To study the requirements of a Railway Reservation System, and to prepare negative test scenarios and test cases that confirm the application handles invalid inputs and wrong user actions safely.

Problem Statement

A railway reservation system lets passengers search trains, check seat availability, book tickets, pay online and cancel tickets. Many users book at the same time, and bookings involve money and limited seats, so a defect can cause wrong bookings, double bookings or refund problems.

Testing is needed to check that the application:

- Passengers can search trains, book tickets and manage bookings correctly.
- Invalid inputs and rule violations are rejected safely.
- Booking limits, payment and cancellation rules follow the requirements.

Any behaviour that differs from the specification must be recorded, with enough detail for developers and testers to repeat and fix it.

Software Requirements

1 Functional requirements (system specification)

ID	Requirement
FR1	The user registers and logs in with a unique user ID and password.
FR2	The user can search trains by source station, destination station and journey date.
FR3	Search results show the train name, number, timings, classes, seat availability and fare.
FR4	A ticket can have up to 6 passengers. Each passenger needs a name, age, gender and an optional berth preference.
FR5	Passenger age must be between 1 and 120, and the name must contain letters only.
FR6	Booking is allowed up to 120 days before the journey date. Past dates are not allowed.
FR7	Payment is possible by UPI, card or net banking within 10 minutes of starting the payment.
FR8	After successful payment, a PNR is generated and the e-ticket is sent by SMS and email.
FR9	When seats are not available, the system books a waiting-list (WL) ticket.
FR10	The user can check the status of a ticket using a 10-digit PNR.
FR11	A ticket can be cancelled before the chart is prepared. The refund follows the cancellation rules.
FR12	Tatkal booking opens at 10:00 AM for AC classes, one day before the journey.

2 Software and environment requirements

Item	Details
Application	Web portal (Java Spring Boot back end, Angular front end)
Database	Oracle / MySQL (users, trains, schedules, bookings, payments)
Operating system	Windows 10/11 or Linux
Browsers	Chrome, Firefox, Edge (latest versions)
Testing tools	Test case document (Excel), Selenium, JMeter, bug tracker (Jira)
Test data	Station codes (MAS, SBC, NDLS), train numbers, passenger details, journey dates, sandbox payment details

Negative Test Scenarios

A negative test scenario checks that the system rejects invalid input or a wrong action, shows a clear message and stays stable and secure.

Scenario ID	Module	Test Scenario	Requirement
RR-NS01	Login	Log in with a wrong password	FR1
RR-NS02	Search	Search with the same source and destination	FR2
RR-NS03	Search	Search with a past journey date	FR6
RR-NS04	Search	Search with an invalid station name	FR2
RR-NS05	Booking window	Book beyond 120 days	FR6
RR-NS06	Booking	Add more than 6 passengers	FR4
RR-NS07	Booking	Enter invalid passenger details	FR5
RR-NS08	Booking	Leave passenger details blank	FR4
RR-NS09	Payment	Payment fails	FR7
RR-NS10	Payment	Payment session expires after 10 minutes	FR7
RR-NS11	PNR status	Check a PNR with an invalid number	FR10
RR-NS12	Cancellation	Cancel a ticket twice	FR11
RR-NS13	Cancellation	Cancel after chart preparation	FR11
RR-NS14	Tatkal	Book Tatkal before the opening time	FR12

Test Cases

Test Case ID	Scenario ID	Test Steps	Test Data	Expected Result
RR-NT01	RR-NS01	1. Enter a valid user ID and a wrong password 2. Click Login	Balaguru01; Wrong@1	Login fails with “Invalid user ID or password”.
RR-NT02	RR-NS02	1. Enter the same station for both 2. Click Find Trains	MAS to MAS	Error “Source and destination cannot be the same” is shown.
RR-NT03	RR-NS03	1. Select yesterday’s date 2. Click Find Trains	Yesterday’s date	Past date is not accepted and an error message is displayed.
RR-NT04	RR-NS04	1. Type an unknown station name 2. Click Find Trains	XXXX	“Station not found” is shown and no results are listed.
RR-NT05	RR-NS05	1. Select a date 121 days from today 2. Search for a train	Date = today + 121 days	Booking is not allowed, and “Booking opens 120 days before the journey” is shown.
RR-NT06	RR-NS06	1. Add 7 passengers to one ticket	7 passengers	The seventh passenger cannot be added and a limit message is displayed.
RR-NT07	RR-NS07	1. Enter age 0, then 121 2. Enter a numeric name 3. Continue	Age 0; age 121; name 1234	Validation errors are shown for age and name. The form does not proceed.
RR-NT08	RR-NS08	1. Click Continue without filling passenger details	Blank	Mandatory-field messages are shown.
RR-NT09	RR-NS09	1. Choose UPI 2. Enter a failing test UPI ID 3. Approve the payment	failure@upi	The ticket is not booked, the seat is released and any debited amount is refunded.
RR-NT10	RR-NS10	1. Go to the payment page 2. Wait for more than 10 minutes 3. Click Pay	Idle for 11 minutes	“Session expired” is shown, and the user must start the booking again.
RR-NT11	RR-NS11	1. Enter a 9-digit PNR, then a PNR with letters 2. Click Check	123456789; ABC1234567	Error “Enter a valid 10-digit PNR” is displayed.
RR-NT12	RR-NS12	1. Cancel a confirmed ticket 2. Try to cancel the same ticket again	Already cancelled ticket	Second cancellation is refused with “Ticket already cancelled”.
RR-NT13	RR-NS13	1. Open a ticket after the chart is prepared 2. Click Cancel	Ticket after chart time	Online cancellation is blocked with a clear message.

Test Case ID	Scenario ID	Test Steps	Test Data	Expected Result
RR-NT14	RR-NS14	1. At 9:30 AM select an AC class 2. Choose Tatkal quota	Time 9:30 AM	Tatkal booking is not available, and the opening time is displayed.

Result / Conclusion

Result: 14 negative test scenarios and 14 test cases were prepared for the Railway Reservation System, covering 9 of the 12 functional requirements (no negative scenario for FR3, FR8, FR9). The test cases by module are:

Module	Test Scenarios	Test Cases
Login	1	1
Search	3	3
Booking window	1	1
Booking	3	3
Payment	2	2
PNR status	1	1
Cancellation	2	2
Tatkal	1	1
Total	14	14

Conclusion: The negative test cases cover invalid inputs, boundary violations, failed payments or operations, and unauthorized actions in the Railway Reservation System. When they are executed on a build, they confirm that the system rejects wrong input safely, shows clear messages, and keeps the data and the user's session secure. Together with the positive test cases, they give a complete set of tests for the stated requirements.